

cs584 s26 - Final Project

Proposal due by: 3/23/2026

Project due by: 4/20/2026

Project Description

The final project is a one- or two-student group project. The required project deliverables are a presentation, software implementation, and a written report. All the components should be submitted together by the due date. Late days are not allowed. To sign up for a topic, make sure that the topic is available and submit a project proposal as described below. There is a 5% penalty for not signing up for a project or not submitting the project proposal on time.

The goal of the project is to deepen your understanding of machine learning or apply it in new applications. You will need to explore recent research papers and select a paper that interests you. Based on the paper you choose, you have two options:

1. **Implement the paper** by writing your own code to execute the concepts discussed. This option strictly prohibits copying or using the source code shared by the paper's authors or any third-party sources.
2. **Modify a significant component of the paper** that you believe can improve the work. Improvement may relate to accuracy, speed, or any other metric you find relevant. Although the aim is to improve upon the original work, your rationale will be more important than the outcome. Successful improvements may earn bonus points. For this option, you can use the author-released source code, but any modifications must be substantial, clearly stated in your proposal, and highlighted with inline comments in the source code.

When selecting a topic, you must identify a paper describing the core approach you will be implementing and the data you will use. The selected paper should contain material different from what was covered in class. Valid project topics include machine learning techniques and applications of machine learning techniques.

Below is a detailed description of the main components of the project.

Project Proposal

The project proposal should contain a short (2 pages) description of the problem you plan to address, the methods you plan to apply, and the data (source) you plan to use. The proposal must rely on one or more recent and published reputable research papers and should include details of these papers. Additionally, include the names of all team members and their respective responsibilities. Responsibilities must be allocated so that issues in the performance of one team member will not affect the other team member,

The proposal and relevant paper(s) should be submitted via Bitbucket, as in the other assignments. Each team member must make a separate submission of the documents. Relevant research papers can be found by searching the online database of the Galvin Library (ACM/IEEE) or using other web resources.

In-Class Presentation (5% of the Final Grade)

To prepare your presentation, follow the research paper you selected in your proposal and, if necessary, other sources. You must clearly indicate the sources used for your presentation and specify their details. Your presentation should be prepared in PowerPoint, HTML, or PDF formats.

Each presentation is allocated approximately 10-15 minutes. In a group project, all group members must present. Time constraints will be strictly enforced to allow sufficient time for other presenters. Planning to use the whiteboard is not allowed. All necessary figures and materials you wish to show during your presentation must be included in your slides. You may record a short video demo, but may not perform a live demo of the program.

Your presentation should include:

- Problem statement and background material.
- Description of the proposed solution.
- Implementation details.
- Discussion of the obtained results.

Part of the presentation grade will be based on peer review. The evaluation criteria will include the following components:

- Facts presented.
- Clarity of the presentation.
- References used.
- Quality of slides.
- Presentation flow.

Report and Software Implementation (15% of the Final Grade)

You need to write a Python program (inside a Python notebook) to implement the algorithms you chose and write a concluding report. Provide instructions on how to use the program you wrote, and supply appropriate test data and configuration files for using it. Ensure that the program allows modification of parameters and the examination of intermediate results.

In writing the program, you may use external modules as needed, provided you cite their origin and clarify that these were not written by you. External modules should not implement the main functionality of your algorithm, which must be written by you. Evaluate the performance of the implemented algorithm and discuss its strengths and weaknesses.

The report should follow the same structure as the presentation. You need to describe the problem you addressed, the method you took to solve it, the data you used, and the results obtained by the program you wrote. You must evaluate and discuss the results. The report should be written in the format of a research paper, be at least 8 pages long, and does not need to follow a two-column format.

The sources used in your work should be clearly indicated in the bibliography section, including web resources and software libraries you used. Cited references should follow IEEE/ACM conventions. At the beginning of the report, include the title of the project, the names of team members, and a **description of what each member accomplished**.

Additional presentation and report requirements

1. Refrain from using code screenshots in your presentation slides or the report. Use instead diagrams to explain network architectures and pseudocode to explain algorithms.
2. Properly evaluating and discussing the results you obtain is very important and will be emphasized when grading.
3. Results should be presented in tables and/or plots with proper sizing and labeling. Do not use too small or too large font sizes.
4. You should evaluate your work/method on more than one metric. For instance, accuracy alone is not sufficient for a classification problem. You should report other performance metrics like f1-score, precision, recall, AUC, etc. Follow the metrics reported in the research paper you selected.
5. When evaluating results, evaluate your own results (not the results from the paper) and explain any differences from the results in the paper. If you fail to obtain reasonable results or unable to implement the project successfully, describe steps you took to solve issues and analyze the reasons behind the failure.

Sign-up and Proposal Instructions

- Find a topic and verify that it is not already taken using the following link:
<https://tinyurl.com/nfatbpxb>
- It is your responsibility to select a topic (and paper) that has not already been chosen by other students. It is acceptable to sign up for a topic selected by another student, but make sure the paper you are selecting is not too similar to the one chosen by the other person.
- Sign up by filling the form using the following link (**Only one form per team.**):
<https://forms.cloud.microsoft/r/Rsnfq3wxm8>
- Prepare a two-page project proposal including the following components, and upload it to Canvas (into a `proposal` assignment). Each team member must submit a copy of the proposal. The proposal should include:
 - Name(s), student ID(s), title of the project.
 - The name of the main paper that will be used and its publication details.
 - Problem statement: What is the problem that needs to be solved?
 - Approach: How will the problem be solved?
 - Data: What data will be used?
 - References: Include relevant papers, web sources, software, and data sources.
 - Team member responsibilities.

Upload the paper(s) you plan to use with the proposal.

- You will receive an email if there are any problems with your proposal. Proposal approvals will be indicated on Canvas.
- To fill the form to sign up for a project, you must be logged into your student account. Pay attention to the instructions at the beginning of the form. If you need to request that we share the document with you, you are not using your student account correctly.

Final Submission Instructions

- Follow the electronic submission instructions provided in the assignments. The project should be submitted as a zip file in the `project` assignment in Canvas.
- In a project with more than one team members, **all team members must submit the entire project.**
- In addition to the `src`, `doc`, and `data` folders, create the following folders:
 - `presentation` : Contains the presentation files and a 4/6 slides-per-page PDF handout.
 - `sources` : Contains any sources you used in the project (e.g., papers and any external code/libraries).
- The report should be provided as a pdf file. Links to the data should be provided in a "data.txt" file, links to saved model files should be provided in a "model.txt" file, and required python modules that need to be installed should be provided in a "requirements.txt" file.
- Do not upload large datasets with your project submission (>10MB). For large datasets, place the data on cloud drive and provide the link in the report.
- Submit a fully executed Python notebook with no gaps in execution. To receive full credit, please ensure the following:
 - The notebook includes all cell outputs and contains no error messages.
 - Re-run your notebook before submitting to ensure that cells are numbered sequentially, starting at [1].
- Late days are not allowed for the final project. If you cannot complete the project by the due date, submit whatever you have available by the deadline.